

This document provides additional assistance with wiring your Extron IP Link Pro Control Processor to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instructions, refer to the user's manual for the specific IP Link Pro Control Processor or the documentation supplied by the manufacturer of the controlled device.

For more information on using Global Scriptor Modules, refer to the "[Guide to Using Scriptor Modules](#)" document.

Device Specifications

Device Type: Video Projector
Manufacturer: Christie
Firmware Version: N/A
Model(s): DHD599-GS, DWU599-GS

Tested on the Following Software and Firmware Versions

IP Link Pro Control Processor Firmware	Global Scriptor Version
2.08.0002-b001	1.4.2

Version History

Module Version	Date	Notes
1_0_2_0	7/30/2018	Adding HDBaseT for Input and PIP Input command.
1_0_1_0	7/10/2018	Fixed Shutter Status
1_0_0_0	1/16/2017	Initial Version

Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'.
Example: `InterfaceName.Unidirectional = 'True'`
- connectionCounter variable must be set to the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15.
Example: `InterfaceName.connectionCounter = 5`

Supported Classes and Examples

SerialClass
<code>InterfaceName = ModuleName.SerialClass(ProcessorName, 'COM1', Model='DHD599-GS')</code>
SerialOverEthernetClass
<code>InterfaceName = ModuleName.SerialOverEthernetClass('192.168.254.254', 2001, Model='DHD599-GS')</code>
EthernetClass
<code>InterfaceName = ModuleName.EthernetClass('192.168.254.254', 3002, Model='DHD599-GS')</code>

Control Commands

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format without Qualifier:

```
InterfaceName.Set(Command, Value)
```

Command AspectRatio	Value 'Auto' 'Letterbox' 'Full Height'	Value 'Native' 'Full Size'	Value '4:3' 'Full Width'
# AspectRatio example InterfaceName.Set('AspectRatio', 'Auto')			
Command AutoImage	Value 'Normal'	Value 'Wide'	
# AutoImage example InterfaceName.Set('AutoImage', 'Normal')			
Command ClosedCaption	Value 'Off'	Value 'Channel 1'	Value 'Channel 2'
# ClosedCaption example InterfaceName.Set('ClosedCaption', 'Off')			
Command Freeze	Value 'On'	Value 'Off'	
# Freeze example InterfaceName.Set('Freeze', 'On')			
Command Input	Value 'VGA' 'DVI-D' 'Presenter' 'DisplayPort'	Value 'Composite' 'Component' 'Card Reader' 'HDBaseT'	Value 'HDMI' 'S-Video' 'Mini USB'
# Input example InterfaceName.Set('Input', 'VGA')			
Command Keypad	Value '0' – '9'		
# Keypad example InterfaceName.Set('Keypad', '0')			
Command LampMode	Value 'Constant Power'	Value 'Constant Intensity'	Value 'Eco Mode'
# LampMode example InterfaceName.Set('LampMode', 'Constant Power')			
Command	Value	Value	Value

Global Scripter Module Communication Sheet

MenuNavigation	'Up'	'Down'	'Left'
	'Right'	'Enter'	'Exit'
	'Menu'		
# MenuNavigation example InterfaceName.Set('MenuNavigation', 'Up')			
Command	Value	Value	
OnScreenDisplay	'On'	'Off'	
# OnScreenDisplay example InterfaceName.Set('OnScreenDisplay', 'On')			
Command	Value	Value	Value
PIPIInput	'VGA'	'Composite'	'HDMI'
	'DVI-D'	'Component'	'S-Video'
	'Presenter'	'Card Reader'	'Mini USB'
	'DisplayPort'	'HDBaseT'	
# PIPInput example InterfaceName.Set('PIPIInput', 'VGA')			
Command	Value	Value	
PIPMODE	'On'	'Off'	
# PIPMode example InterfaceName.Set('PIPMODE', 'On')			
Command	Value	Value	Value
PIPPosition	'Left Vertical Center'	'Top Center'	'Right Vertical Center'
	'Bottom Center'	'Bottom Right'	'Bottom Left'
	'Top Left'	'Top Right'	
# PIPPosition example InterfaceName.Set('PIPPosition', 'Left Vertical Center')			
Command	Value	Value	Value
PIPSize	'Small'	'Medium'	'Large'
# PIPSize example InterfaceName.Set('PIPSize', 'Small')			
Command			
PIPSwap			
# PIPSwap example InterfaceName.Set('PIPSwap', 'Small')			
Command	Value	Value	
Power	'Off'	'On'	
# Power example InterfaceName.Set('Power', 'Off')			
Command	Value	Value	
Shutter	'On'	'Off'	
# Shutter example			

```
InterfaceName.Set('Shutter', 'On')
```

Status Available

For all commands, call Update to receive the latest status. ConnectionStatus does not support the Update function and is triggered by the device providing a successful response to other Update function calls.

Format with Qualifier:

```
InterfaceName.Update(Command, {'Qualifier Key': 'Qualifier Value'})
Value = InterfaceName.ReadStatus(Command, {'Qualifier Key': 'Qualifier Value'})
InterfaceName.SubscribeStatus(Command, {'Qualifier Key': 'Qualifier Value'}, FeedbackHandler)
FeedbackHandler will be called only when the specified qualifier gets a new status.
```

Format without Qualifier:

```
InterfaceName.Update(Command)
Value = InterfaceName.ReadStatus(Command)
InterfaceName.SubscribeStatus(Command, None, FeedbackHandler)
FeedbackHandler will be called when any qualifier gets a new status.
```

Command AspectRatio	Value 'Auto'	Value 'Native'	Value '4:3'
	'Letterbox'	'Full Size'	'Full Width'
	'Full Height'		
# AspectRatio examples InterfaceName.Update('AspectRatio') Value = InterfaceName.ReadStatus('AspectRatio') InterfaceName.SubscribeStatus('AspectRatio', None, FeedbackHandler)			
Command ClosedCaption	Value 'Off'	Value 'Channel 1'	Value 'Channel 2'
# ClosedCaption examples InterfaceName.Update('ClosedCaption') Value = InterfaceName.ReadStatus('ClosedCaption') InterfaceName.SubscribeStatus('ClosedCaption', None, FeedbackHandler)			
Command ConnectionStatus	Value 'Connected'	Value 'Disconnected'	
# ConnectionStatus examples Value = InterfaceName.ReadStatus('ConnectionStatus') InterfaceName.SubscribeStatus('ConnectionStatus', None, FeedbackHandler)			
Command Freeze	Value 'On'	Value 'Off'	
# Freeze examples InterfaceName.Update('Freeze') Value = InterfaceName.ReadStatus('Freeze') InterfaceName.SubscribeStatus('Freeze', None, FeedbackHandler)			
Command Input	Value 'VGA'	Value 'Composite'	Value 'HDMI'

Global Scripter Module Communication Sheet

	'DVI-D'	'Component'	'S-Video'
	'Presenter'	'Card Reader'	'Mini USB'
	'DisplayPort'	'HDBaseT'	
# Input examples InterfaceName.Update('Input') Value = InterfaceName.ReadStatus('Input') InterfaceName.SubscribeStatus('Input', None, FeedbackHandler)			
Command LampMode	Value 'Constant Power'	Value 'Constant Intensity'	Value 'Eco Mode'
# LampMode examples InterfaceName.Update('LampMode') Value = InterfaceName.ReadStatus('LampMode') InterfaceName.SubscribeStatus('LampMode', None, FeedbackHandler)			
Command OnScreenDisplay	Value 'On'	Value 'Off'	
# OnScreenDisplay examples InterfaceName.Update('OnScreenDisplay') Value = InterfaceName.ReadStatus('OnScreenDisplay') InterfaceName.SubscribeStatus('OnScreenDisplay', None, FeedbackHandler)			
Command PIPIInput	Value 'VGA'	Value 'Composite'	Value 'HDMI'
	'DVI-D'	'Component'	'S-Video'
	'Presenter'	'Card Reader'	'Mini USB'
	'DisplayPort'	'HDBaseT'	
# PIPInput examples InterfaceName.Update('PIPIInput') Value = InterfaceName.ReadStatus('PIPIInput') InterfaceName.SubscribeStatus('PIPIInput', None, FeedbackHandler)			
Command PIPMode	Value 'On'	Value 'Off'	
# PIPMode examples InterfaceName.Update('PIPMode') Value = InterfaceName.ReadStatus('PIPMode') InterfaceName.SubscribeStatus('PIPMode', None, FeedbackHandler)			
Command PIPPosition	Value 'Left Vertical Center'	Value 'Top Center'	Value 'Right Vertical Center'
	'Bottom Center'	'Bottom Right'	'Bottom Left'
	'Top Left'	'Top Right'	
# PIPPosition examples InterfaceName.Update('PIPPosition') Value = InterfaceName.ReadStatus('PIPPosition') InterfaceName.SubscribeStatus('PIPPosition', None, FeedbackHandler)			
Command	Value	Value	Value

Global Scripter Module Communication Sheet

PIPSize	'Small'	'Medium'	'Large'
<pre># PIPSize examples InterfaceName.Update('PIPSize') Value = InterfaceName.ReadStatus('PIPSize') InterfaceName.SubscribeStatus('PIPSize', None, FeedbackHandler)</pre>			
Command Power	Value 'Off'	Value 'On'	
<pre># Power examples InterfaceName.Update('Power') Value = InterfaceName.ReadStatus('Power') InterfaceName.SubscribeStatus('Power', None, FeedbackHandler)</pre>			
Command Shutter	Value 'On'	Value 'Off'	
<pre># Shutter examples InterfaceName.Update('Shutter') Value = InterfaceName.ReadStatus('Shutter') InterfaceName.SubscribeStatus('Shutter', None, FeedbackHandler)</pre>			

Cable and Adapter Requirements

Captive Screw to Female DB9 RS-232 Serial Cable

Notes for the Device

To ensure that the projector can be remotely powered on, the Standby Mode option must be set to "Communication Mode" rather than ".5W Mode"

Serial communication

Port Type: RS-232

Baud Rate: 115200

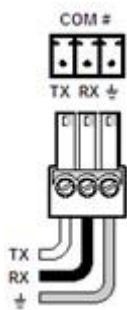
Data Bits: 8

Parity: None

Stop Bits: One

Flow Control: None

Pin Assignments Diagram



Signal	Main Cable	Pin	Signal
TxD		2	TxD
RxD		3	RxD
GND		5	GND



Network communication

When configuring the Ethernet module, be sure device settings match those of the Global Scripter ethernet interface

Port Type:	Ethernet
Default Port:	3002
Logon Credentials Supported:	No
Multi-Connection Capabilities:	Undetermined
Port Changeability:	Undetermined

Ethernet Module Configuration Description

Please refer to user manual for settings and changes to the network communication

Notes for the Device

Appendix A. Set Commands

Aspect Ratio 4:3	(SZP 2)
Aspect Ratio Auto	(SZP 0)
Aspect Ratio Full Height	(SZP 6)
Aspect Ratio Full Size	(SZP 4)
Aspect Ratio Full Width	(SZP 5)
Aspect Ratio Letterbox	(SZP 3)
Aspect Ratio Native	(SZP 1)
Auto Image Normal	(AIM 0)
Auto Image Wide	(AIM 1)
Closed Caption Channel 1	(CLC 1)
Closed Caption Channel 2	(CLC 2)
Closed Caption Off	(CLC 0)
Freeze Off	(FRZ 0)
Freeze On	(FRZ 1)
Input Card Reader	(SIN+MAIN 11)
Input Component	(SIN+MAIN 7)
Input Composite	(SIN+MAIN 9)
Input DVI-D	(SIN+MAIN 5)
Input DisplayPort	(SIN+MAIN 6)
Input HDMI	(SIN+MAIN 4)
Input Mini USB	(SIN+MAIN 12)
Input Presenter	(SIN+MAIN 10)
Input S-Video	(SIN+MAIN 8)
Input VGA	(SIN+MAIN 1)
Input HDBaseT	(SIN+MAIN 13)
Keypad 0	(KEY 36)
Keypad 9	(KEY 34)
Lamp Mode Constant Intensity	(LPM 1)
Lamp Mode Constant Power	(LPM 0)
Lamp Mode Eco Mode	(LPM 2)
Menu Navigation Down	(KEY 42)
Menu Navigation Enter	(KEY 40)
Menu Navigation Exit	(KEY 20)
Menu Navigation Left	(KEY 39)
Menu Navigation Menu	(KEY 19)
Menu Navigation Right	(KEY 41)

Global Scriptor Module Communication Sheet

Revision: 7/30/2018

Menu Navigation Up	(KEY 38)
On Screen Display Off	(OSD 0)
On Screen Display On	(OSD 1)
PIP Input Card Reader	(SIN+PIP 11)
PIP Input Component	(SIN+PIP 7)
PIP Input Composite	(SIN+PIP 9)
PIP Input DVI-D	(SIN+PIP 5)
PIP Input DisplayPort	(SIN+PIP 6)
PIP Input HDMI	(SIN+PIP 4)
PIP Input Mini USB	(SIN+PIP 12)
PIP Input Presenter	(SIN+PIP 10)
PIP Input S-Video	(SIN+PIP 8)
PIP Input VGA	(SIN+PIP 1)
PIP Mode Off	(PIP 0)
PIP Mode On	(PIP 1)
PIP Position Bottom Center	(PPP 3)
PIP Position Bottom Left	(PPP 5)
PIP Position Bottom Right	(PPP 4)
PIP Position Left Vertical Center	(PPP 0)
PIP Position Right Vertical Center	(PPP 2)
PIP Position Top Center	(PPP 1)
PIP Position Top Left	(PPP 6)
PIP Position Top Right	(PPP 7)
PIP Size Large	(PHS 2)
PIP Size Medium	(PHS 1)
PIP Size Small	(PHS 0)
PIP Swap None	(PPS)
Power Off	(PWR 0)
Power On	(PWR 1)
Shutter Off	(SHU 0)
Shutter On	(SHU 1)

Appendix B. Update Commands

Aspect Ratio	(SZP?)
Closed Caption	(CLC?)
Freeze	(FRZ?)
Input	(SIN+MAIN?)
Lamp Mode	(LPM?)
On Screen Display	(OSD?)
PIP Input	(SIN+PIP?)
PIP Mode	(PIP?)
PIP Position	(PPP?)
PIP Size	(PHS?)
Shutter	(SHU?)